

Static-analysis engineer at ISP RAS / SharpChecker with professional LLVM experience from MCST. Public projects demonstrate Roslyn ControlFlowGraph fixed-point data flow across loops/joins and conservative interprocedural LLVM analysis with a separate legality phase.

EXPERIENCE

- 2026 — present

ISP RAS — Static Analysis Engineer

 - Contribute to SharpChecker, ISP RAS's industrial static-analysis platform for C#.NET.
- July — August 2026

MCST — Compiler Engineering Intern

 - Developed loop/interprocedural analyses used to decide transformation legality rather than assuming unsupported cases are safe.
 - Built a conservative Global-IV prototype with APInt affine reasoning and transitive call-effect analysis before LLVM IR mutation.

PROJECTS

- DerefAfterNullAnalyzer

A Roslyn ControlFlowGraph analyzer propagates null state over conditional edges to a fixed point, handles loops/back edges, and merges states from multiple predecessors.

Handles method/property/index/event dereferences and invalidation after assignment/ref/out; dedicated analyzer tests use Microsoft.CodeAnalysis.CSharp.Testing.
- LLVM Interprocedural Global IV Optimization

An LLVM 22 pass localizes supported induction-like globals after interprocedural effects/affine analysis and a separate legality decision; APInt models fixed-width wraparound.

29 positive/negative regression cases: global-iv,verify, structural expectations, idempotence, and observable before/after execution comparison.
- UniversalToolchain

Modular compiler/runtime infrastructure: typed artifact contracts, dependencies/conflicts, provider selection, pass ordering, and artifact routes compile into an immutable LanguagePlan before execution.

Current exact test manifest: 1,324/1,324 passing; interpreter/CIL parity, exact package binding, and fail-closed runtime checks protect the selected execution plan.

SKILLS

- Static / Program Analysis

C#.NET, Roslyn ControlFlowGraph, fixed-point data flow, conditional edges, loops/back edges, joins, state invalidation
- Compiler / IR

C++23, LLVM 22/IR, CFG/SSA, APInt affine reasoning, interprocedural effects, legality
- Correctness / Testing

worklist convergence, positive/negative regression cases, LLVM Verifier, unsupported-case rejection
- Supporting Infrastructure

typed contracts, immutable plans, interpreter/CIL parity, Linux, Git

EDUCATION

HSE University — Software Engineering, 2026–2030

RECOGNITION

- LangDev'26 — Build the Language, Then Make the Abstractions Disappear

Accepted talk on UniversalToolchain's architecture; LangDev'26 takes place October 8–9, 2026 in Málaga.
- UniversalToolchain — Baltic Science and Engineering Competition

First Degree Diploma and Grand Prize "Perfection as Hope" for UniversalToolchain.
- MEPhI Junior

Overall winner; First Degree Diploma, 96/100 for UniversalToolchain.

Moscow / remote